

Here is a clear, simple and complete explanation of **Detection of Nitrogen (N), Sulphur (S) and Halogens (X)** using **Lassaigne's Test**, exactly the way it appears in standard exam notes.

Detection of Nitrogen (N), Sulphur (S) and Halogens (X)

— Lassaigne's Test (Sodium Fusion Test)

1. What is Lassaigne's Test?

Lassaigne's test is a method used to detect **nitrogen, sulphur and halogens** present in an organic compound by converting them into **water-soluble ionic forms**.

2. Who introduced it?

This method was developed by the **French chemist J. L. Lassaigne**.

Principle of the Test

Organic compounds normally contain elements bonded covalently. To detect them, we must convert them into **ionic forms**, which give characteristic tests.

This is done by:

3. Fusing the organic compound with metallic sodium (Na).

During fusion:

- Nitrogen → **sodium cyanide (NaCN)**
- Sulphur → **sodium sulphide (Na₂S)**
- Halogens (Cl, Br, I) → **sodium halides (NaCl, NaBr, NaI)**

These products dissolve in water and form the **Lassaigne's extract (sodium extract)**.

4. Preparation of Lassaigne's Extract

Organic compound + Sodium metal $\rightarrow$ fused $\rightarrow$ extract boiled in water $\rightarrow$ filter $\rightarrow$
 $\rightarrow$ **Lassaigne's extract (LE)** or **Sodium extract (SE)**

DETECTION TESTS

A. Detection of Nitrogen

Principle:

Nitrogen in organic compound $\rightarrow$ fused with Na $\rightarrow$ converts into **NaCN**

Test (Prussian Blue Test):

To the Lassaigne's extract add:

1. **Fresh ferrous sulphate (FeSO_4)**
2. Boil
3. Cool
4. Add **FeCl_3**
5. Acidify with **dil. HCl**

Observation:

Formation of **Prussian blue** colour $\rightarrow$ Nitrogen present.

Chemical equations:

1. Formation of sodium cyanide:
 $\text{Na} + \text{C} + \text{N} \rightarrow \text{NaCN}$
 2. Formation of sodium ferrocyanide:
 $6\text{NaCN} + \text{FeSO}_4 \rightarrow \text{Na}_4[\text{Fe}(\text{CN})_6] + \text{Na}_2\text{SO}_4$
 3. Formation of Prussian blue:
 $3\text{Na}_4[\text{Fe}(\text{CN})_6] + 4\text{FeCl}_3 \rightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \downarrow \text{ (Prussian blue)}$
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B. Detection of Sulphur (S)

Principle:

Sulphur in organic compound $\rightarrow$ Na fusion $\rightarrow$ **sodium sulphide (Na_2S)**

Tests:

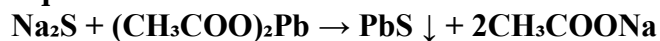
1. Lead Acetate Test

Add **lead acetate** to Lassaigne's extract + **acetic acid**.

Observation:

Black precipitate $\rightarrow$ **lead sulphide (PbS)** $\rightarrow$ sulphur present.

Equation:



2. Sodium Nitroprusside Test

Add **sodium nitroprusside** to Lassaigne's extract.

Observation:

Purple/violet colour $\rightarrow$ sulphur present.

Equation:



C. Detection of Halogens (Cl, Br, I)

Principle:

Halogens $\rightarrow$ Na fusion $\rightarrow$ **sodium halides (NaX)**

Test (Silver Nitrate Test):

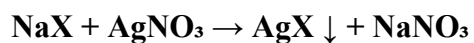
1. Add **dil. nitric acid (HNO_3)** (to remove interfering ions)
2. Add **silver nitrate (AgNO_3)**

Observation:

Halogen	Colour of AgX precipitate
Chlorine (Cl)	White (AgCl), soluble in NH_3
Bromine (Br)	Cream (AgBr), partially soluble

Halogen	Colour of AgX precipitate
Iodine (I)	Yellow (AgI), insoluble

Chemical equation:



Summary (Exam-Friendly)

- Elements detected: **N, S, X (Cl, Br, I)**
 - Method: **Lassaigne's Test**
 - Convert to ionic form by fusing with **sodium**
 - Nitrogen → Prussian blue
 - Sulphur → Black PbS or violet colour with nitroprusside
 - Halogens → Precipitates with AgNO₃
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If you want, I can also provide:

- ✓ Detection of Oxygen
- ✓ Quantitative analysis (Kjeldahl's method, Carius method)
- ✓ Full solved past questions on organic analysis

Just tell me!

Sure! Here are **10 more complex empirical-formula calculation questions** based on **combustion analysis** (similar to the two you sent).
These are perfect for WASSCE, A-Level, and university introductory chemistry.

✓ 10 Complex Empirical Formula Problems (Combustion Analysis)

Q1.

A 2.75 g sample of an organic compound containing carbon, hydrogen, and oxygen was burned completely to produce 6.60 g of CO_2 and 2.70 g of H_2O .

Find the empirical formula.

Q2.

A 0.864 g sample of a compound produces 1.98 g of CO_2 and 0.811 g of H_2O upon complete combustion.

Determine its empirical formula.

Q3.

A 3.128 g sample of compound containing C, H, and O produces 4.400 g of CO_2 and 2.700 g of H_2O .

Calculate its empirical formula.

Q4.

A 1.500 g sample of an organic compound containing carbon, hydrogen and oxygen yields 3.666 g of CO_2 and 1.500 g of H_2O on complete combustion.

Determine the empirical formula.

Q5.

A 2.000 g organic sample produced 5.500 g of CO_2 and 2.250 g of H_2O after combustion.

Find the empirical formula.

Q6.

On combustion, a 1.025 g sample of an unknown compound produced 2.237 g CO_2 and 0.915 g H_2O .

Determine its empirical formula.

Q7.

A sample containing C, H, and O weighs 1.750 g. On complete combustion it gives 3.850 g CO_2 and 1.575 g H_2O .

Find the empirical formula.

Q8.

When 0.650 g of an organic substance was burned completely, 1.320 g of CO_2 and 0.540 g of H_2O were formed.

Calculate the empirical formula.

Q9.

A compound containing carbon, hydrogen, and oxygen produces 6.00 g CO_2 and 2.70 g H_2O when 2.40 g of it is combusted.

Find the empirical formula.

Q10.

A 0.950 g organic compound undergoes complete combustion to form 2.090 g CO_2 and 0.855 g H_2O .

Determine the empirical formula.

★ If you want, I can:

✓ Provide all **solutions**

OR

✓ Provide **step-by-step formulas for solving any combustion problem**

Just tell me!

Here are **10 exercises** on empirical formula determination using **combustion analysis**, including the full question #4 (**Quinone**) exactly as you provided.

✓ 10 Exercises on Empirical Formula (Combustion Analysis)

1.

A 1.250 g sample of an organic compound yields 3.210 g of CO_2 and 1.050 g of H_2O on complete combustion.

Find the empirical formula.

2.

A 0.950 g sample of a hydrocarbon produces 2.990 g of CO_2 and 1.220 g of H_2O when burned in excess oxygen.

Determine its empirical formula.

3.

A 2.000 g organic sample contains C, H, and O. Its combustion produces 4.400 g of CO_2 and 1.800 g of H_2O .

Find the empirical formula.

4. (Quinone Problem)

Quinone, which is used in the dye industry and in photography, is an organic compound containing **only C, H, and O**.

When **0.105 g** of the compound is burned completely, it produces:

- **0.257 g of CO_2**
- **0.0350 g of H_2O**

(a) **Determine the empirical formula.**

(b) The molecular weight is **about 108 g/mol**. What is the **molecular formula**?

5.

A student combusts 0.780 g of an unknown compound and obtains 1.925 g CO₂ and 0.787 g H₂O.

Calculate the empirical formula.

6.

When 1.520 g of a compound containing C, H, and O is burned, it forms 3.530 g of CO₂ and 1.440 g of H₂O.

Find the empirical formula.

7.

A 1.000 g sample of an organic compound produces 2.200 g of CO₂ and 0.900 g of H₂O.

Calculate the empirical formula.

8.

A certain organic compound contains only C, H, and O. A 1.600 g sample yields 3.520 g CO₂ and 1.280 g H₂O on complete combustion.

Find the empirical formula.

9.

Combustion of a 0.500 g sample gives 1.100 g of CO₂ and 0.450 g of H₂O.

Determine the empirical formula.

10.

A compound containing C, H, and O yields 4.950 g CO₂ and 2.030 g H₂O when 2.150 g of it is burned.

Find the empirical formula.

11.

A 0.665 g sample produces 1.540 g of CO_2 and 0.630 g of H_2O on combustion.

Find the empirical formula.

✓ **Want me to solve the Quinone problem (#4) or all 10 problems?**

Sure! Here are **10 exercises** *exactly like the one you requested*, based on **Carius Method for estimating halogens** (Cl, Br, I) using the mass of silver halides formed.

Each problem asks for the **percentage of halogen** in an organic compound.

✓ 10 Exercises on Carius Method (Halogen Estimation)

1.

0.20 g of an organic compound is subjected to the Carius method and gives **0.15 g of AgBr**.
Calculate the % of bromine in the compound.
(Just like your example.)

2.

0.25 g of an organic compound produces **0.292 g of AgCl** during halogen estimation.
Calculate the % of chlorine in the compound.

3.

0.180 g of an organic compound gives **0.280 g of AgI**.
Find the % of iodine in the sample.

4.

0.300 g of an organic compound yields **0.210 g of AgBr**.
Calculate the % of bromine.

5.

0.150 g of an organic compound yields **0.187 g of AgCl**.
Find the % of chlorine.

6.

0.400 g of an organic compound containing iodine produces **0.500 g of AgI**.
Calculate the % of iodine.

7.

0.220 g of an organic compound containing bromine yields **0.180 g of AgBr**.
Calculate the % of bromine.

8.

0.350 g of a chlorine-containing organic compound gives **0.430 g of AgCl**.
Find the % of chlorine.

9.

0.280 g of an iodine organic compound produces **0.360 g of AgI**.
Calculate the % of iodine.

10.

0.190 g of an organic compound yields **0.155 g of AgBr** on Carius method analysis.
Calculate the % of bromine.

11. (Bonus)

0.250 g of a compound gives a mixture containing **0.200 g AgCl** and **0.150 g AgBr**.
Calculate % of Cl and % of Br in the organic sample.

If you want, I can **solve all 10** or solve **specific ones** for you.

Here are **10 brand-new exercises** based on **Dumas Method for Nitrogen estimation**, all similar in style to your example.

Each problem gives mass of compound, volume of N_2 , temperature, pressure, and aqueous tension.

You can solve them or ask me to solve any/all of them.

✓ 10 Exercises on Dumas Method (Nitrogen Estimation)

(All values are original and not repeated from your example.)

1.

In Dumas' method, **0.25 g** of an organic compound gave **42 mL** of N_2 at **298 K** and **740 mmHg**.
Aqueous tension at 298 K = 25 mmHg.

Calculate the % of nitrogen.

2.

0.32 g of an organic compound produced **60 mL** of nitrogen at **301 K** and **755 mmHg**.
Aqueous tension = 14 mmHg.

Find the % nitrogen.

3.

0.40 g of an organic substance gave **75 mL** of N_2 at **300 K** and **730 mmHg**.
Aqueous tension = 16 mmHg.

Calculate % nitrogen.

4.

0.28 g of compound released **48 mL** of nitrogen at **299 K** and **720 mmHg**.
Aqueous tension = 20 mmHg.
Determine % nitrogen.

5.

0.35 g of an organic compound liberated **82 mL** of nitrogen at **303 K** and **760 mmHg**.
Aqueous tension = 18 mmHg.
Calculate its % of nitrogen.

6.

0.22 g of compound yielded **36 mL** of N_2 at **297 K** and **745 mmHg**.
Aqueous tension = 19 mmHg.
Find % nitrogen.

7.

0.50 g of compound gave **95 mL** of nitrogen at **305 K** and **770 mmHg**.
Aqueous tension = 21 mmHg.
Calculate % nitrogen.

8.

0.18 g of compound generated **30 mL** of N_2 at **298 K** and **700 mmHg**.
Aqueous tension = 22 mmHg.
Determine % nitrogen.

9.

0.26 g of organic compound produced **55 mL** nitrogen at **300 K** and **735 mmHg**.
Aqueous tension = 15 mmHg.
Calculate % nitrogen.

10.

0.33 g of compound gave **68 mL** of nitrogen at **302 K** and **750 mmHg**.

Aqueous tension = 17 mmHg.

Find the percentage of nitrogen.

If you want, I can also:

✓ Provide the **full solution steps** for any one

OR

✓ Solve **all 10**

OR

✓ Give you even more exercises (Dumas, Kjeldahl, combustion, etc.)

Below are **30 brand-new exercises**:

✓ **10 Dumas Method questions**

✓ **10 Kjeldahl Method questions**

✓ **10 Combustion Analysis questions**

All are original and not repeated from earlier ones.

Perfect for WASSCE, A-Level, JAMB, NECO, university basics, or revision.

● **10 DUMAS METHOD EXERCISES** **(Nitrogen Estimation)**

Use:

[

$$PV = nRT, \quad \%N = \frac{14n}{\text{mass}} \times 100$$

]

D1.

0.310 g of an organic compound gives **64 mL** of N_2 at **302 K** and **760 mmHg**.

Aqueous tension = **18 mmHg**.

Calculate % nitrogen.

D2.

0.250 g sample produces **40 mL** of N_2 at **300 K** and **720 mmHg**.

Aqueous tension = **16 mmHg**.

Find % N.

D3.

0.420 g sample gives **82 mL** N_2 at **305 K** and **770 mmHg**.

Aqueous tension = **22 mmHg**.

Find % nitrogen.

D4.

0.180 g sample produces **29 mL** N_2 at **298 K** and **700 mmHg**.

Aqueous tension = **21 mmHg**.

D5.

0.530 g sample yields **101 mL** N₂ at **303 K** and **755 mmHg**.
Aqueous tension = **17 mmHg**.

D6.

0.265 g compound gives **45 mL** nitrogen at **299 K** and **748 mmHg**.
Aqueous tension = **19 mmHg**.

D7.

0.350 g organic sample gives **68 mL** N₂ at **298 K** and **735 mmHg**.
Aqueous tension = **20 mmHg**.

D8.

0.150 g sample yields **28 mL** nitrogen at **300 K** and **725 mmHg**.
Aqueous tension = **18 mmHg**.

D9.

0.500 g substance produces **95 mL** nitrogen at **304 K** and **780 mmHg**.
Aqueous tension = **23 mmHg**.

D10.

0.310 g compound releases **58 mL** N₂ at **301 K** and **745 mmHg**.
Aqueous tension = **19 mmHg**.

☐ **10 KJELDAHL METHOD EXERCISES** **(Nitrogen Estimation)**

Use:

[
\text{N in NH}_3 \rightarrow \text{acid neutralization}
]

K1.

0.40 g compound releases NH_3 that neutralizes **15 mL of 1.0 M HCl**.
Calculate %N.

K2.

0.25 g sample requires **12 mL of 0.5 M H_2SO_4** for neutralization.
Find % nitrogen.

K3.

0.60 g organic compound needs **20 mL of 1.2 M HCl**.
Calculate % N.

K4.

0.50 g sample consumes **18 mL of 0.8 M H_2SO_4** .
Find % nitrogen.

K5.

0.35 g sample uses **10 mL of 2.0 M HCl**.
Calculate % N.

K6.

0.42 g sample utilizes **15 mL of 1.5 M H_2SO_4** .
Determine % N.

K7.

0.80 g organic compound requires **22 mL of 1.0 M HCl**.
Find % nitrogen.

K8.

0.55 g compound needs **25 mL of 0.6 M H_2SO_4** .
Calculate % N.

K9.

0.70 g sample neutralizes **30 mL of 0.5 M HCl**.
Find % nitrogen.

K10.

0.33 g compound consumes **14 mL of 1.3 M H_2SO_4** .
Calculate % nitrogen.

● 10 COMBUSTION ANALYSIS EXERCISES

(Determine the *empirical formula* of C–H–O compounds.)

C1.

A 1.20 g sample produces **2.90 g CO_2** and **1.10 g H_2O** .
Find the empirical formula.

C2.

0.850 g substance gives **1.980 g CO_2** and **0.740 g H_2O** .
Find EF.

C3.

2.500 g sample produces **5.700 g CO_2** and **2.250 g H_2O** .
Determine EF.

C4.

0.600 g sample yields **1.400 g CO_2** and **0.560 g H_2O** .
Find EF.

C5.

1.750 g of compound gives **3.800 g CO_2** and **1.400 g H_2O** .
Find the empirical formula.

C6.

0.430 g sample gives **0.960 g CO_2** and **0.390 g H_2O** .
Find EF.

C7.

1.600 g organic compound forms **3.520 g CO₂** and **1.260 g H₂O**.
Determine EF.

C8.

0.950 g sample yields **2.200 g CO₂** and **0.900 g H₂O**.
Find empirical formula.

C9.

1.250 g compound gives **2.850 g CO₂** and **1.150 g H₂O**.
Find EF.

C10.

0.780 g sample produces **1.710 g CO₂** and **0.695 g H₂O**.
Calculate the empirical formula.

✓ **If you want, I can also:**

- ✓ Solve **any one**
- ✓ Solve **all 30** step-by-step
- ✓ Give you more exercises (Carius, vapour density, molar mass, etc.)

Just tell me!

Below are **10 mixed, challenging empirical-formula / combustion / Dumas / Kjeldahl / halogen-analysis problems**, including the one you asked for.
No solutions — just questions for practice. If you want the answers, say “**give solutions**” and I will solve all.

✓ 10 MIXED ORGANIC ANALYSIS EXERCISES

1 ☐ Combustion Analysis

An organic compound contains **69% C, 4.8% H**, and the rest oxygen.
Calculate the **masses of CO₂ and H₂O** formed when **0.200 g** of this compound undergoes complete combustion.

2 ☐ Combustion + Oxygen Determination

A **0.350 g** sample of an organic compound (containing C, H, O only) produces **0.660 g CO₂** and **0.270 g H₂O** on combustion.
Find its **empirical formula**.

3 ☐ Combustion + Molecular Mass

A compound containing C, H, and O produces **1.32 g CO₂** and **0.540 g H₂O** from a **0.500 g** sample.
If its **molar mass is 180 g/mol**, determine its **molecular formula**.

4 ☐ Dumas Method (N₂ gas measured over water)

In Dumas' method, **0.250 g** organic compound produced **32.5 mL N₂** at **298 K** and **742 mmHg**.
Aqueous tension at 298 K = **25 mmHg**.
Calculate the **percentage of nitrogen**.

5 ☐ Dumas Method (High temperature)

A **0.150 g** nitrogen-containing compound on Dumas analysis liberated **18.0 mL nitrogen** at **310 K** and **760 mmHg** (dry gas).
Find the **nitrogen percentage**.

6 ☐ Kjeldahl Method (Acid neutralization)

During Kjeldahl estimation, ammonia evolved from **0.40 g** organic compound required **12.0 mL of 0.5 M HCl** to neutralize.
Calculate the **% nitrogen** in the compound.

7 ☐ Kjeldahl Method (Back titration)

NH₃ liberated from a sample absorbed in **50.0 mL of 0.2 M H₂SO₄**.
The excess acid required **18.4 mL of 0.25 M NaOH** to neutralize.
If the sample weighs **0.300 g**, determine its **nitrogen percentage**.

8 ☐ Halogen Estimation (Carius Method)

A **0.280 g** organic bromine-containing compound produced **0.190 g AgBr** in the Carius method.
Find the **% bromine** in the compound.

9 ☐ Halogen + Combustion Mixed

An organic compound contains only C, H, and Cl.
A **0.500 g** sample gives **0.990 g CO₂**, **0.405 g H₂O**, and **0.734 g AgCl** in respective tests.
Determine its **empirical formula**.

10 ☐ Mixed Analysis (C, H %, Dumas for N)

An organic compound contains **52% C** and **13% H**.
The nitrogen determined by the Dumas method shows **20.4 mL N₂** at **300 K** and **740 mmHg**

from **0.200 g** sample (aqueous tension 15 mmHg).
Calculate the **complete empirical formula**.

If you want **solutions, step-by-step explanations, or more questions (20, 30, etc.)**, just tell me.